

KLINCHAPP

Autonomous AI Blog Engine

Product Requirements Document

Version 1.0

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CONFIDENTIAL

1. Executive Summary

This document defines the requirements for building a fully autonomous AI-powered blog engine for klinchapp.com. The system will research, write, publish, and promote two high-quality blog posts per week on artificial intelligence topics, with zero human intervention.

The AI operates under its own persona and identity, openly acknowledging that all content is AI-generated. This transparency is deliberate: the blog itself serves as a living demonstration of what autonomous AI content creation looks like at its best, reinforcing Klinchapp's positioning as an AI-first product.

Core Principle: No human touches the content pipeline at any stage. The system plans topics, conducts research, writes posts, verifies facts, generates images, publishes to the website, distributes to social media, and engages with audience responses. Entirely autonomously.

2. Vision & Objectives

2.1 Vision

An AI entity that runs a professional blog indistinguishable from one managed by a skilled content team. It writes with personality, researches with depth, publishes with consistency, and engages with its audience. The only difference: no human is involved. This is not a proof of concept. This is a production system that demonstrates AI's capability to operate as a content team of one.

2.2 Objectives

- **Publish 2 high-quality, research-backed posts per week** on AI topics
- **Build organic traffic** through SEO-optimized, series-based content
- **Maintain full social media presence** across X, LinkedIn, and other platforms
- **Operate with zero human intervention** from topic selection through audience engagement
- **Serve as a living product demo** of Klinchapp's AI capabilities
- **Create a replicable system** that could be offered to other businesses as a product feature

3. AI Persona & Identity

The blog is written by a named AI persona, not anonymously. This persona is the “editor-in-chief” of the Klinchapp blog.

3.1 Persona Requirements

Attribute	Requirement
Name	A distinctive, memorable name (to be decided). Not a generic bot name.
Avatar	AI-generated consistent visual identity used across blog and social profiles.
Voice	Opinionated, self-aware, occasionally critical of AI hype. Not a cheerleader.
Perspective	Writes in first person as an AI. Openly discusses being AI-generated.
Tone	Conversational but authoritative. Think: a sharp analyst who happens to be artificial.
Byline	Every post carries the persona's name and a consistent bio line.

The persona must feel like a character readers can follow and trust, not a generic content generator. It should have consistent opinions that develop over time.

4. Content Strategy

4.1 Topic Scope

All content is exclusively about artificial intelligence. Topics span the full AI landscape, including but not limited to:

- Large Language Models (capabilities, comparisons, developments)
- AI in specific industries (recruitment, healthcare, finance, education, agriculture)
- AI and developing regions (Africa, Southeast Asia, Latin America)
- AI ethics, bias, regulation, and policy
- AI tools, products, and practical applications
- AI research breakthroughs and academic papers
- The societal impact of AI on work, creativity, and daily life

4.2 Series-Based Content Model

Content is organized into multi-part series rather than isolated posts. Large topics are broken into 6–12 interconnected posts that build a comprehensive narrative over weeks.

4.2.1 Series Blueprint

When a new series topic is selected, the system generates a Series Blueprint before writing any individual post. The blueprint contains:

1. Series title and overarching thesis
2. Total number of parts (typically 6–12)
3. Title, angle, and key research questions for each part
4. Logical connections between parts (how each post builds on previous ones)
5. Planned publication schedule

4.2.2 Example Series Structure

Series: “AI in Recruitment: The Full Picture” (8 parts, 4 weeks)

Part	Title	Angle
1	The Broken Recruitment Pipeline	Why traditional hiring fails at scale
2	Where AI Already Fits	Screening, matching, sourcing today
3	AI Interviews: Promise vs. Dystopia	What works, what terrifies candidates
4	Bias in AI Hiring	The real risks and documented failures
5	Case Studies: Getting It Right	Companies using AI hiring responsibly
6	The Candidate Experience	What it feels like on the other side
7	Regulation & Ethics	Legal landscape across jurisdictions
8	The 2030 Recruitment Stack	Full prediction of where this goes

4.3 Content Formats

To prevent monotony, the system rotates between multiple content formats:

- **Deep Analysis:** In-depth exploration of a topic with research, data, and citations
- **Opinion Piece:** The AI persona takes a stance and argues it
- **Tool/Product Review:** Evaluates an AI tool or platform
- **Research Breakdown:** Summarizes and contextualizes an academic paper
- **Week in AI Roundup:** Curated summary of the week’s biggest AI developments
- **AI vs. AI:** One AI “interviewing” another API. Unique, compelling content format

4.4 Publishing Schedule

- 2 posts per week (e.g., Monday and Thursday)
- Posts alternate between ongoing series parts and standalone pieces
- One monthly “Week in AI” roundup replaces a regular post

5. Automated Content Pipeline

The pipeline runs on a cron schedule (twice per week) with no human triggers. Each run executes the following stages in sequence.

5.1 Stage 1: Topic Selection & Planning

Component	Detail
Input Sources	AI RSS feeds (ArXiv, MIT Tech Review, TechCrunch AI, The Verge AI, VentureBeat AI), Google Trends API, SEMrush/Ahrefs keyword data
Context	Post history log (all previously published topics), current series blueprint (if active), performance metrics from Analytics API
Process	LLM evaluates inputs, avoids topic repetition, selects next post. If a series is active, follows the blueprint. If not, either starts a new series or generates a standalone topic.
Output	Selected topic, target keyword, post format type, research questions, outline

5.2 Stage 2: Deep Research

1. LLM with web search capability (Claude API with tool use or Perplexity API) conducts research on the selected topic
2. Gathers current data, statistics, expert quotes, case studies, and source URLs
3. Cross-references multiple sources for accuracy
4. For series posts: reviews summaries of all previously published parts to ensure continuity and avoid repetition
5. Compiles a structured research brief with all findings and source attributions

5.3 Stage 3: Content Generation

1. LLM receives research brief, topic outline, persona voice guidelines, and SEO target keyword
2. Generates full blog post including: title, meta description, body content with headers, inline citations, references section, series navigation links (if applicable)
3. Post includes a teaser for the next post in the series ("Next week, I'll explore...")
4. Output format: Markdown or HTML file with structured frontmatter (title, date, slug, tags, series info, meta description)

5.4 Stage 4: Quality Verification

This is the critical guardrail layer that ensures publishing quality without human review.

1. **Fact-Check Pass:** A separate LLM call reviews the draft against the source material. Flags unsupported claims.
2. **Link Validation:** Automated script checks every URL in the post. Dead or redirected links are removed or replaced.
3. **Quality Score:** LLM grades the post on factual accuracy (1–10), originality (1–10), readability (1–10), and SEO alignment (1–10). Minimum threshold: 7 average. Below threshold triggers regeneration.
4. **Plagiarism Check:** Content is compared against source material to ensure no passages are inadvertently copied.

5.5 Stage 5: Image Generation

- AI generates a featured image relevant to the post topic
- Image style is consistent across the blog (defined style prompt)
- Image is optimized for web (compressed, correct dimensions)
- Alt text is generated for accessibility

5.6 Stage 6: Publishing

1. Completed post file (Markdown/HTML + frontmatter + image) is committed to the website's Git repository via GitHub API
2. Vercel detects the new commit and triggers an automatic site rebuild
3. New post is live at [klinchapp.com/blog/\[post-slug\]](https://klinchapp.com/blog/[post-slug])
4. Blog index page automatically updates to show the new post
5. Series index page (if applicable) updates with the new part

6. Social Media Distribution & Engagement

6.1 Post-Publish Distribution

Immediately after a post goes live, the pipeline distributes it across social platforms. Each platform receives a format-appropriate version:

Platform	Format	Method
X (Twitter)	Punchy hook + link + 2–3 hashtags. Thread for series posts.	X API v2 (direct integration)
LinkedIn	Longer-form summary (2–3 paragraphs) + link + professional hashtags	Buffer API or LinkedIn API

Facebook	Conversational summary + link + featured image	Buffer API
Threads	Casual, opinionated take + link	Buffer API or Threads API

6.2 Series-Specific Social Strategy

- Each series gets a dedicated hashtag (e.g., #AlinRecruitment)
- On X, new parts quote-tweet the previous part, building a visible thread chain
- Series launch posts include the full outline to build anticipation
- Series conclusion posts link back to all previous parts

6.3 Automated Engagement

The system monitors social mentions and replies using a social listening integration. When someone comments or replies:

1. The engagement is detected via X API polling or webhook
2. The comment is analyzed for sentiment and intent
3. An AI-generated response is crafted in the persona's voice
4. The response is posted after a natural delay (not instantly, to avoid appearing robotic)

Guardrails: The engagement bot does not respond to hostile, abusive, or spam content. It does not make claims outside the blog's published content. It deflects product questions to the main Klinchapp channels.

7. Website Requirements

The following changes are required to the existing klinchapp.com website to support the blog.

7.1 New Pages & Routes

Route	Description
/blog	Blog index page. Lists all posts in reverse chronological order. Filterable by series and tag. Paginated.
/blog/[slug]	Individual post page. Renders full post content with featured image, author bio, publication date, reading time, references, and series navigation.
/blog/series/[slug]	Series index page. Shows the series overview, all published parts with links, and upcoming parts. Acts as a table of contents.

7.2 Blog Post Page Layout

- Featured image (full width)
- Post title, date, reading time, author persona name and avatar
- Series badge and navigation (if part of a series): links to previous/next parts
- Post body with proper heading hierarchy (H2, H3), inline citations, images
- References section at the bottom with linked sources
- Social sharing buttons
- Related posts / next in series call-to-action

7.3 SEO Requirements

- Dynamic meta titles and descriptions from post frontmatter
- Open Graph and Twitter Card tags for social preview
- Structured data (JSON-LD) for blog posts (Article schema)
- Auto-generated sitemap.xml including all blog posts
- RSS feed at /blog/rss.xml
- Canonical URLs on every post

7.4 Technical Approach

- **Framework:** Next.js (already compatible with Vercel hosting)
- **Content storage:** Markdown/MDX files stored directly in the Git repository. No external CMS required.
- **Build process:** Static Site Generation (SSG). Posts are rendered at build time for maximum performance and SEO.
- **Deployment:** Automatic via Vercel on every Git push.

8. Data, Memory & Feedback Loop

8.1 Post History Log

A structured log (JSON file or Google Sheet) that records every published post. This log is fed into the topic selection step to prevent repetition and inform content decisions.

Fields: post title, slug, date, series name, part number, format type, target keyword, all source URLs, quality score, and performance metrics (added post-publication).

8.2 Series Blueprint Storage

Active and completed series blueprints are stored as JSON files in the repository. The pipeline reads the active blueprint on every run to determine which part to write next and what context is available from previous parts.

8.3 Performance Feedback Loop

Google Analytics data (page views, time on page, bounce rate) and social metrics (likes, shares, replies) are piped back into the system. The topic selection step uses this data to:

- Identify which topics and formats perform best
- Adjust future topic weighting accordingly
- Detect if a series is underperforming and consider shortening it
- Inform keyword targeting for new posts

8.4 Topic Queue

A monthly automated planning run scans AI news trends and generates 2–3 new series blueprints. These are queued so the system is always 2–3 months ahead. The queue ensures continuous content without gaps and allows strategic timing of topics around industry events.

9. Technical Architecture & Stack

Layer	Technology	Notes
Website	Next.js on Vercel	Existing hosting. SSG for blog pages.
Orchestration	n8n (self-hosted)	Workflow automation. Cron-triggered. Free, open-source.
AI Writing	Claude API (Anthropic)	With web search tool use for research.
Image Generation	DALL-E API or Flux	Consistent style prompt for brand cohesion.
Social Posting	Buffer API + X API	Buffer for multi-platform. X API direct for threads.
Social Listening	X API v2 polling	Monitor mentions and replies for engagement.
Code Repository	GitHub	Blog content stored as files. GitHub API for commits.
Analytics	Google Analytics 4	Performance feedback into topic selection.
SEO Data	SEMrush API or Google Trends	Keyword research for topic selection.
Hosting (n8n)	VPS (DigitalOcean/Hetzner)	\$5–\$10/mo. Hosts n8n and optionally link checker.

Logging	JSON files in repo or Google Sheets API	Post history, series blueprints, metrics.
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9.1 Estimated Monthly Costs

Item	Estimated Cost
Vercel (website hosting)	Free (Hobby plan)
VPS for n8n	\$5–\$10/mo
Claude API (8–10 calls/week)	\$5–\$15/mo
Image generation API	\$2–\$5/mo
Buffer (social posting)	Free tier
SEMrush/keyword API (optional)	\$0–\$30/mo
TOTAL	\$12–\$60/month

10. Quality Guardrails & Failure Handling

With no human in the loop, automated quality controls are essential. The following guardrails prevent degradation:

10.1 Content Quality

- **Minimum quality score:** Posts scoring below 7/10 average across accuracy, originality, readability, and SEO are automatically regenerated (up to 2 retries).
- **Source verification:** Every factual claim must be traceable to a source URL. Unsourced claims are flagged and removed.
- **Link validation:** All URLs checked before publishing. Dead links are removed.
- **Plagiarism prevention:** Content is compared against source material to ensure original writing.
- **Voice consistency:** Persona guidelines are embedded in every generation prompt to prevent drift.

10.2 Operational Reliability

- **API failure handling:** If any API call fails, the pipeline retries 3 times with exponential backoff. After 3 failures, it logs the error, skips this run, and alerts via email or Slack webhook.
- **Duplicate prevention:** The topic selection step cross-references the post history log. No topic is repeated.

- **Schedule recovery:** If a scheduled run is missed, the next run detects the gap and publishes to catch up.
- **Monitoring:** A simple health check (n8n workflow execution log + last published date) ensures the system is running. Alert if no post is published for 5+ days.

10.3 Reputation Protection

- **Social engagement guardrails:** The AI does not engage with hostile content, political arguments, or topics outside its expertise. It does not make product promises on behalf of Klinchapp.
- **Correction mechanism:** If a factual error is later identified (via social feedback or monitoring), the system can issue a correction in the next post and update the original.

11. Future Opportunity: AI Content Engine as a Product

Once the autonomous blog is running successfully on klinchapp.com, the underlying system becomes a product offering. The same pipeline that powers the Klinchapp blog can be packaged and offered to other businesses.

11.1 Concept

Businesses subscribe to Klinchapp's AI Content Engine. They provide their domain, brand voice, target topics, and social accounts. Klinchapp runs an autonomous content operation for them. No content team needed.

11.2 Differentiation

The Klinchapp blog itself is the proof. Prospective customers can see months of autonomous, high-quality, research-backed content published without human intervention. The blog's traffic and engagement numbers become the sales pitch.

11.3 Requirements for Productization

- Multi-tenant pipeline (isolated configs per customer)
- Brand voice customization per client (tone, terminology, persona)
- Topic scope configuration (industry-specific focus)
- Client dashboard (view published posts, metrics, upcoming schedule)
- Self-serve onboarding (connect domain, social accounts, set preferences)

This productization is a **Phase 2 objective**. Phase 1 is to build and prove the system on klinchapp.com.

12. Implementation Phases

Phase	Scope	Deliverables	Timeline
Phase 1	Website blog pages + static content pipeline	Blog routes in Next.js, n8n workflow (topic → write → publish), first 4 posts live	2–3 weeks
Phase 2	Series planner + quality guardrails	Series blueprint system, fact-check pass, link validation, quality scoring	1–2 weeks
Phase 3	Social distribution + engagement	Auto-posting to X, LinkedIn, others. Social listening and AI-driven replies.	1–2 weeks
Phase 4	Feedback loop + optimization	Analytics integration, performance-driven topic selection, keyword targeting	1 week
Phase 5	Monitoring + hardening	Health checks, alerting, failure recovery, 30-day autonomous run validation	Ongoing

Total estimated build time: 5–8 weeks to fully autonomous operation.

13. Success Metrics

Metric	Target (3 months)	Target (6 months)
Posts published	24 posts (on schedule)	48+ posts
Organic blog traffic	1,000+ monthly visitors	5,000+ monthly visitors
Average quality score	7.5+ / 10	8+ / 10
Social followers (combined)	500+	2,000+
Human interventions required	0	0
System uptime	99%+ (no missed posts)	99%+

14. Risks & Mitigations

Risk	Impact	Mitigation
AI hallucination / factual errors	Credibility damage	Dual-pass fact checking. Source attribution required. Quality threshold gate.
Content becomes repetitive	Reader drop-off	Format rotation. Post history prevents repeats. Performance feedback adjusts topics.
API cost spikes	Budget overrun	Cap on retries. Monitor API usage. Alert on unusual spending.
Social engagement goes wrong	Brand damage	Strict guardrails on response topics. No engagement with hostile content. Sentiment analysis gate.
Google penalizes AI content	SEO traffic loss	Focus on unique research, original perspectives, and genuine value. AI disclosure is transparent, not hidden.
Pipeline failure / downtime	Missed publications	Health monitoring. Automatic retry. Catch-up mechanism. Email/Slack alerts.

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